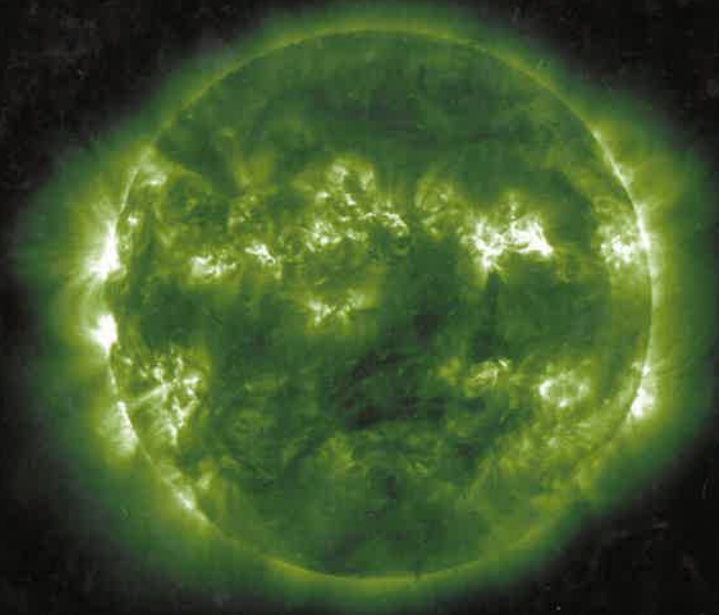


Lecture-Tutorials

FOR INTRODUCTORY ASTRONOMY



Edward E. Prather, Timothy F. Slater, Jeffrey P. Adams, Gina Brissenden

THIRD EDITION

Lecture-Tutorials

FOR INTRODUCTORY ASTRONOMY

THIRD EDITION

Edward E. Prather
University of Arizona

Timothy F. Slater
University of Wyoming

Jeffrey P. Adams
Millersville University

Gina Brissenden
University of Arizona

with contributions by

Jack A. Dostal
Wake Forest University

Colin S. Wallace
University of Arizona

PEARSON

Boston Columbus Indianapolis New York San Francisco Upper Saddle River
Amsterdam Cape Town Dubai London Madrid Milan Munich Paris Montreal Toronto
Delhi Mexico City São Paulo Sydney Hong Kong Seoul Singapore Taipei Tokyo

In the celestial sphere model, Earth is stationary and the stars are carried on a sphere that rotates about an axis that points at the North Star. In Figure 1 below, two stars, A and B, are each shown at four different positions (1, 2, 3, and 4) through which each star will pass during the course of one revolution of the celestial sphere. In addition, your location on Earth in the Northern Hemisphere and the portion of the celestial sphere that is above your horizon are shown.

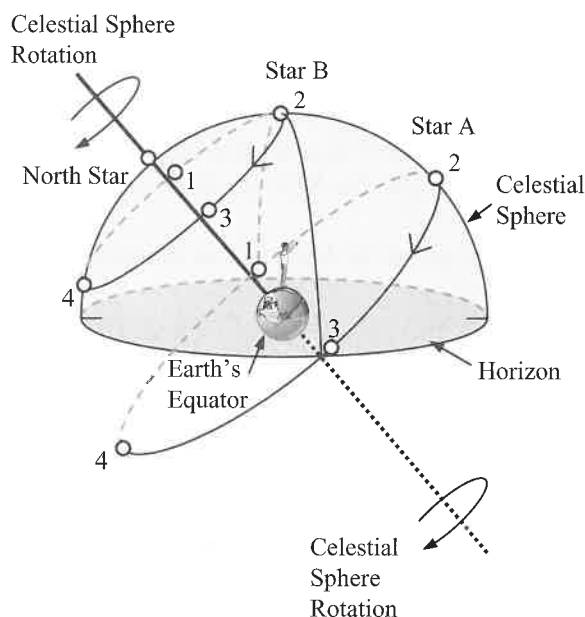


Figure 1

- 1) Is the horizon shown a real physical horizon or an imaginary plane that extends from your observing location on Earth out to the stars?
- 2) Can the observer shown see a star when it is located below the horizon? Why or why not?
- 3) Is either Star A or B ever in an unobservable position? If so, which position(s)?
- 4) When a star travels from a position below the observer's horizon to a position above the observer's horizon, is that star rising or setting?
- 5) When a star travels from a position above the observer's horizon to a position below the observer's horizon, is that star rising or setting?

- 6) Star A is just visible above your eastern horizon at Position 1. At which of the numbered positions is it just visible above your western horizon?
- 7) At which position(s), if any, does Star B rise and set?
- 8) Two students are discussing their answers to Question 7.

Student 1: *Locations B1 and B3 are on my horizon because they are rising and setting just like A1 and A3.*

Student 2: *Figure 1 shows that Star B is as low as it will get when it is just above the northern horizon at B4. So Star B never goes below the horizon.*

Do you agree or disagree with either or both of the students? Explain your reasoning.

- 9) Label the directions north, south, east, and west on Figure 1. Check your answer with another group.
- 10) For each indicated position, describe where in the sky you must look to see the star at that time. Each description requires two pieces of information: the direction you must face (north, northeast, east, etc.) and how far above the horizon you must look (low, high, or directly overhead). If you cannot see the star, state that explicitly. The descriptions for four positions are given as examples.
- a) A1: *east, low*
 - b) A2:
 - c) A3:
 - d) A4:
 - e) North Star: *north, high*
 - f) B1:
 - g) B2: *directly overhead*
 - h) B3: *northwest, high*
 - i) B4:

Check your answers with a nearby group and resolve any inconsistencies.

- 11) Does Star B ever set?

- 5) In Figure 1, the path that Star B follows is shown with a dashed line. Draw a small arrowhead on the path to represent the direction Star B would be moving at the instant it is at each of the four locations marked with an **x**. Check your answers with a nearby group.
- 6) Using Figure 2, describe in words where you would look to see Star A when it is halfway between rising and setting.

Part II: Looking East

Figure 3 shows an extended view along the eastern horizon showing the positions of Stars A and B at 6 P.M. The arrow shown is provided to indicate the direction that Star B will be moving at 6 P.M.

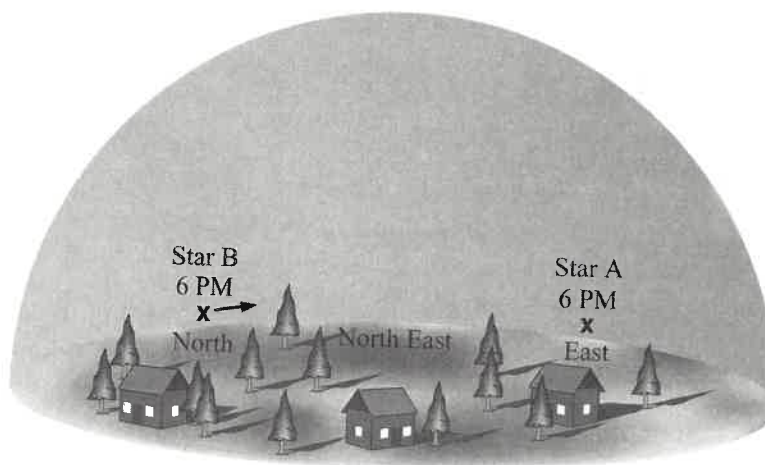


Figure 3

- 7) Recall that in Question 6, you found that Star A ends up high in the southern sky when it is halfway between rising and setting (and therefore never passes through your zenith). Draw a straight arrow at the **x** in the east in Figure 3 (the position of Star A at 6 P.M.) to indicate the direction Star A moves as it rises. Studying Figure 2 can also help clarify your answer.
- 8) Two students are discussing the direction of motion of a star rising directly in the east.
- Student 1:** Stars move east to west so any star rising directly in the east must be moving straight up so that it can end up in the west. If the arrow were angled, the star would not set in the west.
- Student 2:** I disagree. From Figure 2, the path of Star A starts in the east, then it moves high in the southern sky yet still sets in the west. To do this it has to move toward the south as it rises so I drew my arrow angled up and to the right.
- Do you agree or disagree with either or both of the students? Explain your reasoning.

9) Imagine you could see Star B at noon. Fifteen minutes later, in what direction will Star B have moved? Explain your reasoning.

10) Consider the student comment below.

Student: *The amount of time that all stars are above the horizon is 12 hours because it takes 12 hours for a star to rise in the east and then set in the west.*

Do you agree or disagree with the student? Explain your reasoning.

Consider the situation shown below in which the Sun and a group of constellations are shown at sunrise, Figure 4, and then shown again 8 hours later, Figure 5.

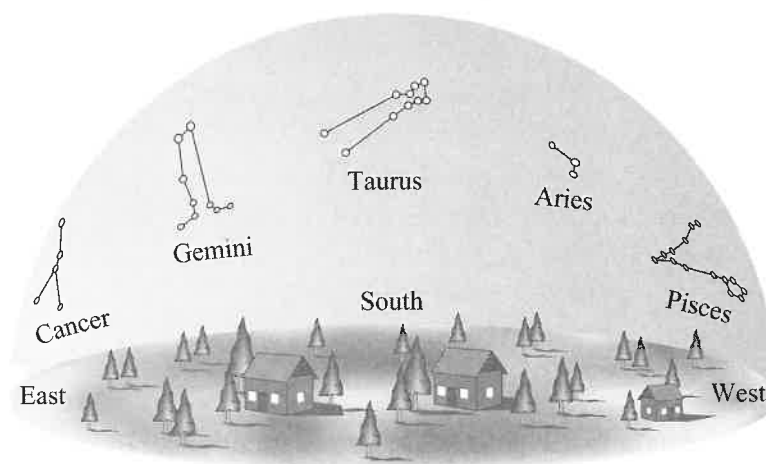


Figure 4

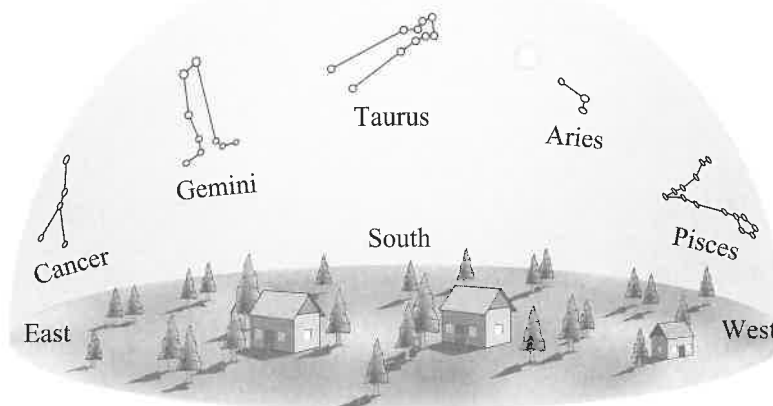


Figure 5

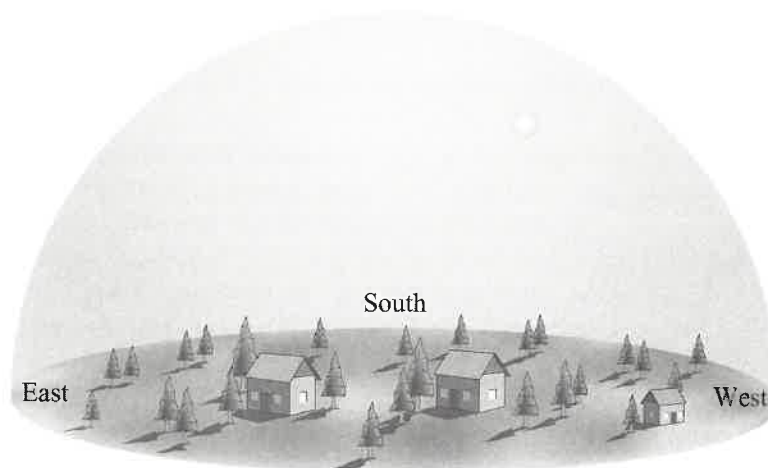
- 11) Consider the following debate between two students regarding the motion of the Sun and constellations shown in Figures 4 and 5.

Student 1: *We know the Sun rises in the east and moves through the southern part of the sky and then sets in the west. Eight hours after sunrise, it makes sense that the Sun will have moved from being on the eastern horizon near the constellation Cancer to being located high in the southwestern sky near the constellation Aries.*

Student 2: *You're forgetting that some stars and constellations also move from the east through the southern sky and to the west just like the Sun. So, the Sun will still be near Cancer eight hours later. So Figure 5 is drawn incorrectly. It should show that the constellations have all moved like the Sun, so Cancer would also be located high in the southwestern sky, with the Sun, eight hours later.*

Do you agree or disagree with either or both of the students? Explain your reasoning. Check your answers with another group.

- 12) In Question 11, we found that Figure 5 was drawn incorrectly. Redraw Figure 5 on the figure below by sketching the approximate location of any constellations from Figure 5 that would still be visible.



Part I: Monthly Differences

Figure 1 shows a Sun-centered, or heliocentric, perspective view of the Earth–Sun system indicating the direction of both the daily rotation of Earth about its own axis and its annual orbit about the Sun. You are the observer shown in Figure 1, located on Earth in the Northern Hemisphere while facing south.

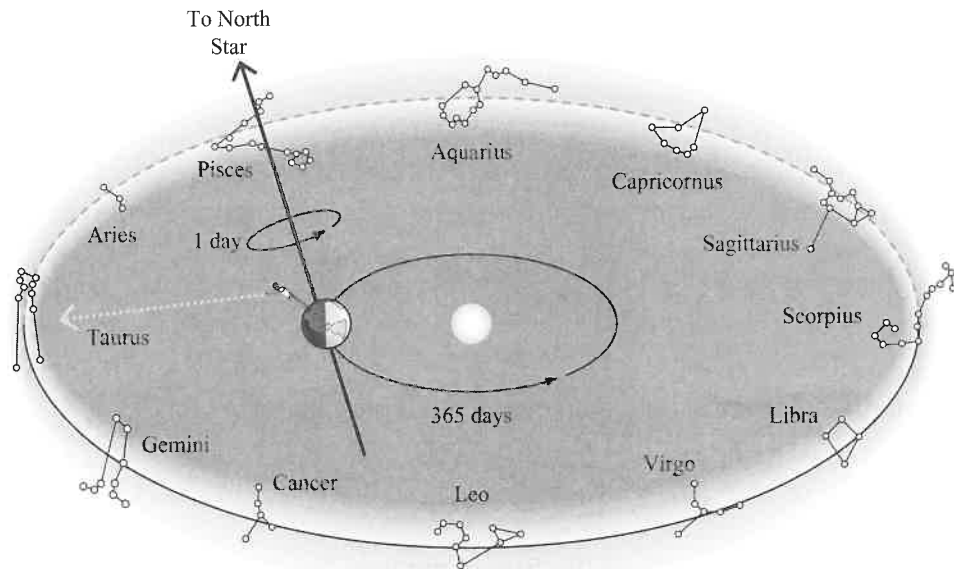


Figure 1

Figure 2 shows a horizon view of what you would see when facing south on this night at the same time as shown in Figure 1.

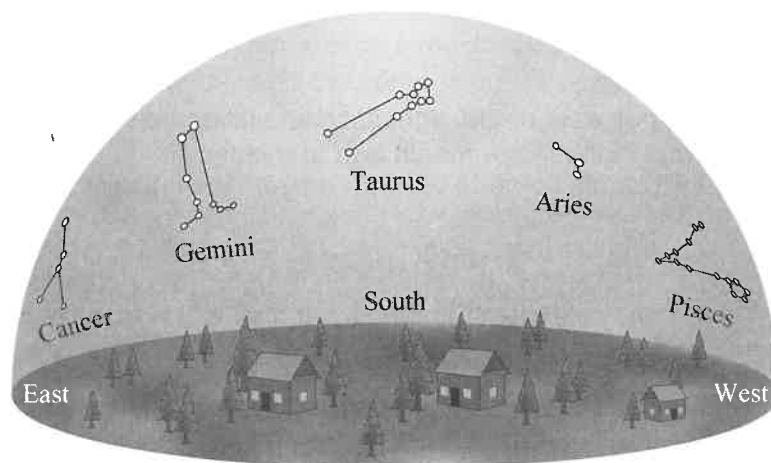


Figure 2

- 1) Which labeled constellation do you see highest in the southern sky?
- 2) For the time shown, which constellation is just to the east (i.e., to your left when you are facing south) and which constellation is just to the west (i.e., to your right when you are facing south) of the highest constellation at this instant?

east:

west:

- 3) Noting that you are exactly on the opposite side of Earth from the Sun, what time is it?
- 4) In six hours, will the observer be able to see the Sun? If not, why not? If so, in what direction (north, south, east, or west) would you look to see the Sun?
- 5) Which constellation will be behind the Sun at the time described in Question 4?
- 6) When it is noon for the observer, which constellation will be behind the Sun?
- 7) One month later, Earth will have moved one-twelfth of the way around the Sun. You are again facing south while observing at midnight. Which constellation will now be highest in the southern sky?
- 8) Do you have to look east or west of the highest constellation that you see now to see the constellation that was highest one month ago at midnight?
- 9) Does the constellation that was highest in the sky at midnight a month ago now rise earlier or later than it rose last month? Explain your reasoning.

Part II: Daily Differences

Figure 3 shows the same Earth–Sun view as before and the bright star Betelgeuse, which is between Taurus and Gemini.

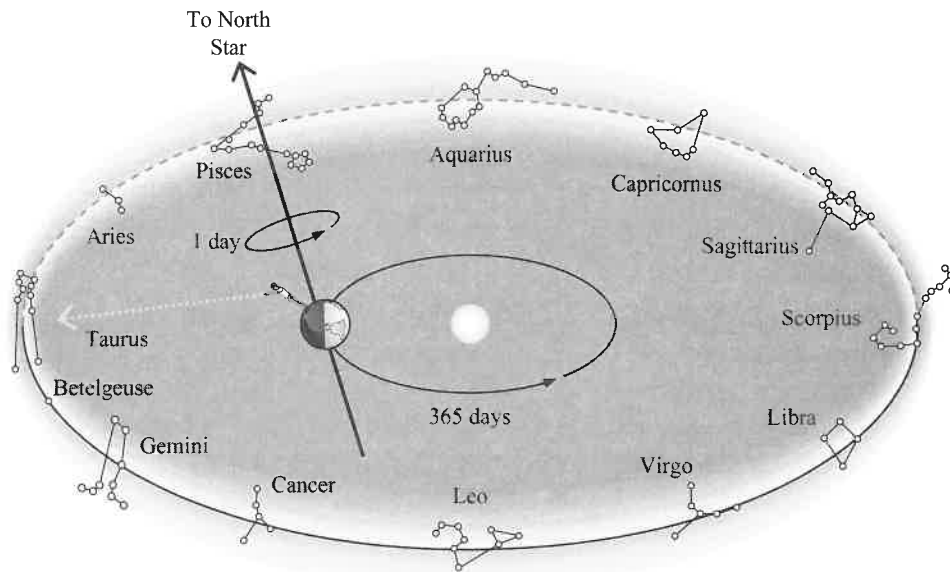


Figure 3

10) Imagine last night you saw the star Betelgeuse just starting to rise on your eastern horizon at 7:15 P.M. At 7:15 P.M. tonight, will Betelgeuse be above, below, or exactly on your eastern horizon?

11) Two students are discussing their answers to Question 10.

Student 1: *Earth makes one complete rotation about its axis each day so Betelgeuse will rise at the same time every night. It will therefore be exactly on the eastern horizon.*

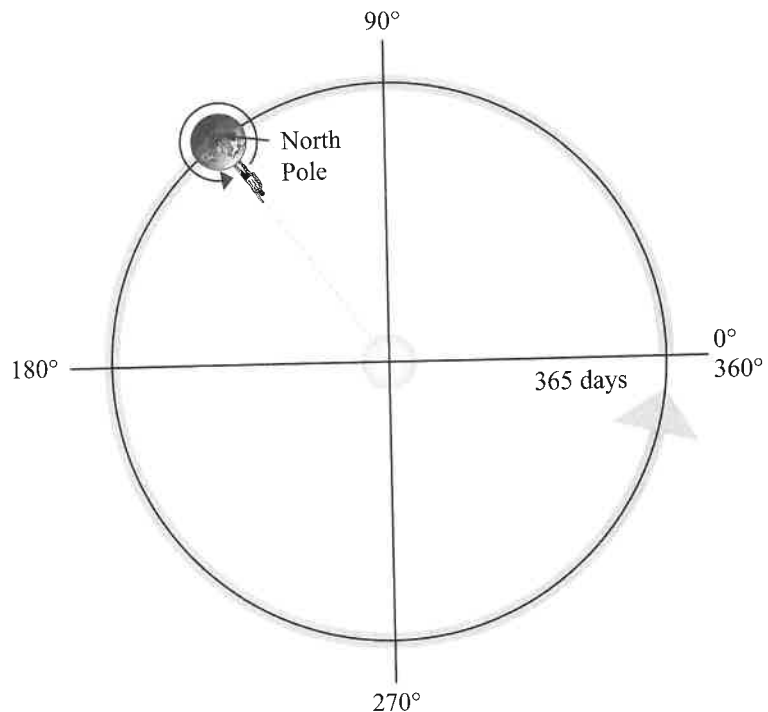
Student 2: *No. Because Earth goes around the Sun, the constellation Taurus rises earlier each month and so it must rise a little bit earlier each night, too. Betelgeuse must do the same thing. Tonight it would rise a little before 7:15 and be above the eastern horizon by 7:15.*

Do you agree or disagree with either or both of the students? Explain your reasoning.

Part I: Solar Day

Figure 1 shows a top-down view of the Earth–Sun system. Arrows indicate the directions of the rotational and orbital motions of Earth. For the observer shown, the Sun is highest in the sky at noon.

- 1) Earth orbits the Sun in a counterclockwise direction once every 365 days. Approximately how many degrees does Earth move along its orbit in one day?

**Figure 1**

- 2) As Earth orbits the Sun, it also rotates in a counterclockwise direction about its axis as shown in Figure 1. We define 24 hours as the time from when the Sun is highest in the sky one day to when it is highest in the sky the next day. How many degrees does Earth rotate about its axis in exactly 24 hours: 360° , slightly less than 360° , or slightly more than 360° ?
- 3) How long does it take Earth to rotate exactly 360° : slightly less than 24 hours, 24 hours, or slightly more than 24 hours?
- 4) Two students are discussing their answers to Questions 2 and 3.

Student 1: *Earth rotates about its axis once every 24 hours, and one rotation equals 360° .*

Student 2: *No. When Earth has gone around 360° it has also moved a small amount counterclockwise around the Sun, which means the Sun is not yet at its highest point. Earth must spin a little bit more for the Sun to reach its highest point.*

Do you agree or disagree with either or both of the students? Explain your reasoning.

Part II: Sidereal Day

We define a **solar day** as the time it takes for the Sun to go from its highest point in the sky on one day to its highest point in the sky on the next day, and we divide that time into 24 hours.

A **sidereal day** is defined as the time it takes for Earth to rotate *exactly* 360° about its axis with respect to the distant stars.

- 5) When does Earth rotate a greater amount, during a solar day or during a sidereal day?
- 6) Which takes a shorter amount of time, a solar day or a sidereal day?

Note: Since Earth rotates more than 360° in a solar day, a sidereal day is about 4 minutes shorter than a solar day.

Imagine that at some time in the future the direction that Earth orbits the Sun is somehow reversed so that Earth now orbits the Sun approximately 1° *clockwise* each day. However, the rotation about its own axis remains counterclockwise at the same rate.

- 7) In the space below, create a sketch similar to Figure 1 to depict this imaginary situation.
- 8) Through how many degrees will Earth now rotate in a *sidereal* day?
- 9) Through how many degrees will Earth now rotate in a *solar* day?
- 10) Which is now longer, the solar or the sidereal day?
- 11) Is a sidereal day now longer, shorter, or the same length as a sidereal day was before we changed Earth's orbital direction?
- 12) Is a solar day now longer, shorter, or the same length as a solar day was before we changed Earth's orbital direction?

For all parts of this activity, it is helpful to imagine that the stars are so bright (or our Sun so dim) that the stars can be seen during the day so that your sky might appear as in Figure 1.

Part I: Daily Motion

On December 1, at noon, you are looking toward the south and see the Sun among the stars of the constellation Scorpius as shown in Figure 1.

- 1) At 3 P.M. that afternoon, will the Sun appear among the stars of the constellation Capricornus, Sagittarius, Scorpius, Libra, or Virgo?

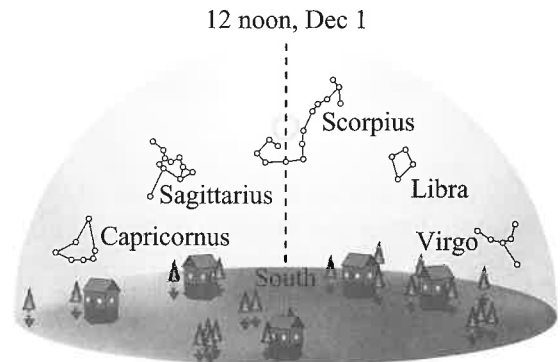


Figure 1

- 2) Two students are discussing their answers to Question 1.

Student 1: *The Sun moves from the east through the southern part of the sky and then to the west. By 3 P.M. it will have moved from being high in the southern sky to the west into the constellation Libra.*

Student 2: *You're forgetting that stars and constellations, like those in Figure 1, will rise in the east, move through the southern sky, and then set in the west just like the Sun. So the Sun will still be in Scorpius at 3 P.M.*

Do you agree or disagree with either or both of the students? Explain your reasoning.

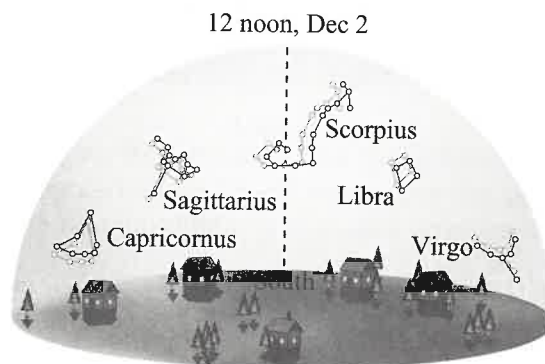
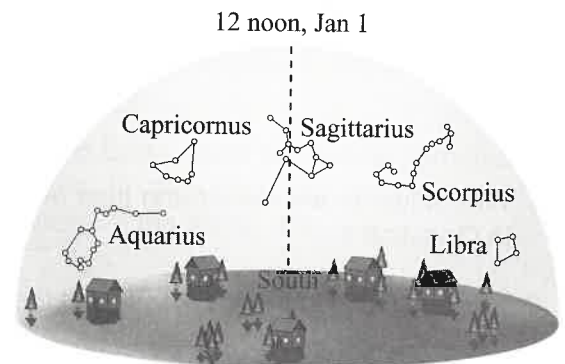
Recall that in the celestial sphere model, the stars' daily motions result from the rotation of the celestial sphere.

- 3) Is it reasonable to account for the Sun's **daily motion** by assuming that the Sun is at a fixed position on the celestial sphere (in this case in the location of the constellation Scorpius) and is carried along its path across the sky by the sphere's rotation? Explain why or why not.

Part II: Monthly Changes

By careful observation of the Sun's position in the sky throughout the year, we find that the celestial sphere rotates slightly more than 360° every 24 hours. Figure 2 shows the same view of the sky (as Figure 1) but on December 2 at noon. For comparison, the view from the previous day at the same time is also shown in gray.

- 4) Draw the location of the Sun as accurately as possible in Figure 2.
- 5) Figure 3 shows the same view of the sky (as Figure 1) one month later on January 1 at noon. Draw the location of the Sun as accurately as possible in this figure.

**Figure 2****Figure 3**

- 6) Two students are discussing their answers to Questions 4 and 5.

Student 1: *The Sun will always lie along the dotted line in the figures when it's noon.*

Student 2: *But, we saw in Question 3 that the Sun's motion can be modeled by assuming it is stuck to the celestial sphere. The Sun must, therefore, stay in Scorpius.*

Student 1: *If that were true, then by March the Sun would be setting at noon. The Sun must shift a little along the celestial sphere each day so that in 30 days it has moved toward the east into the next constellation.*

Do you agree or disagree with either or both of the students? Explain your reasoning.

- 7) Why is it reasonable to think of the Sun as attached to the celestial sphere over the course of a single day, as suggested in Question 3, even though we know from Questions 5 and 6 that the Sun's position is not truly fixed on the celestial sphere?

Part III: The Ecliptic

The zodiacal constellations were of special interest to ancient astronomers because these are the constellations through which the Sun moves throughout the year. This was incorporated into their celestial sphere model by having the Sun loosely fixed to the celestial sphere but allowing it to slip a small amount each day. The Sun's position on the celestial sphere (among the stars in the constellation Scorpius) on December 1 is shown in Figure 4.

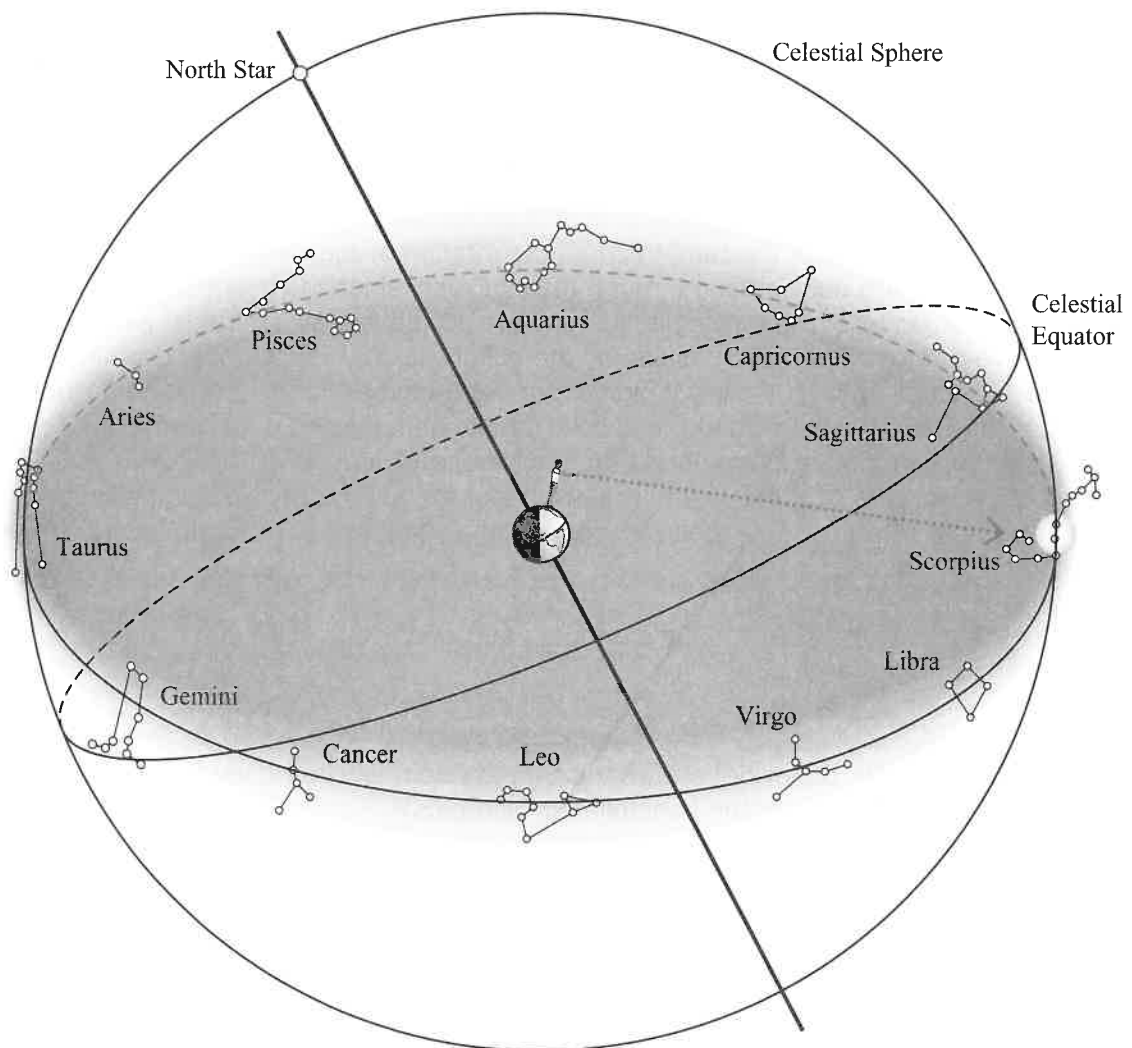


Figure 4

- 8) On Figure 4, draw where the Sun will be located on the celestial sphere on January 1. Label this position "Jan 1."
- 9) On Figure 4, for the other constellations, draw in the Sun and label the constellation with the approximate date that the Sun will be located there.

The line drawn through these constellations, tracing out the Sun's annual path, is called the **ecliptic**.

- 10) Label the ecliptic in Figure 4.

- 11) About how many times does the celestial sphere rotate in the time it takes the Sun to move between two adjacent constellations (i.e., 1/12 of the way around) along the ecliptic?

- 12) How long does it take the Sun to make one complete trip around the ecliptic (i.e., from Scorpius to Scorpius)?

Part IV: Wrap-Up

It is important to realize that the ecliptic represents an *annual* drift of the Sun and does not represent the daily path of the Sun. Instead, the rotation of the celestial sphere is responsible for the Sun's daily motion through the sky. Also, since the ecliptic is tilted with respect to the rotation axis of the celestial sphere, the ecliptic slowly "wobbles" as the celestial sphere rotates. The Sun's position on the ecliptic is only important in deciding whether the Sun's daily path will carry it high in the sky (summer) or low in the sky (winter). In Figure 5a, the Sun's position along the ecliptic and its path for one day (dashed line) are shown for two different dates: December 1 (in Scorpius) and June 1 (in Taurus). Figures 5b, 5c, and 5d show the path of the Sun and the wobble of the ecliptic at six-hour intervals as the celestial sphere rotates. Study these figures, carefully noting that the ecliptic and Sun are both carried by the celestial sphere.

- 13) On Figure 5d, label the ecliptic (Sun's annual path) and the Sun's daily path for December 1 and June 1.
- 14) Which Figure (5a, 5b, 5c, or 5d) shows the Sun at noon, low in the southern sky, when it would be among the stars of the constellation Scorpius?
- 15) Which Figure (5a, 5b, 5c, or 5d) shows the Sun at noon, high in the southern sky, when it would be among the stars of the constellation Taurus?

December 1 (Sun in Scorpius)
and
June 1 (Sun in Taurus)

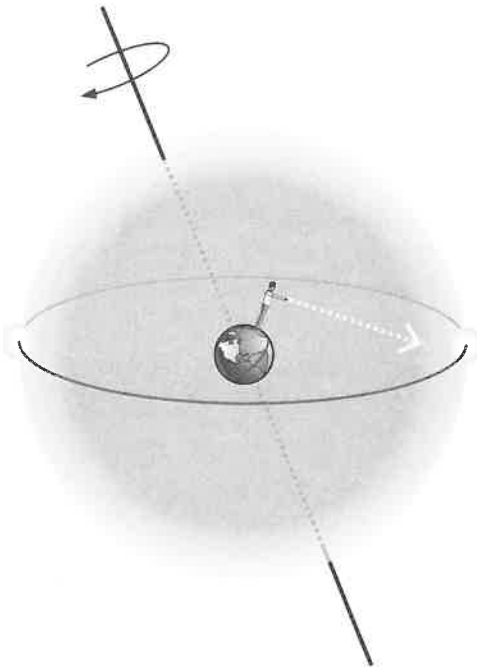
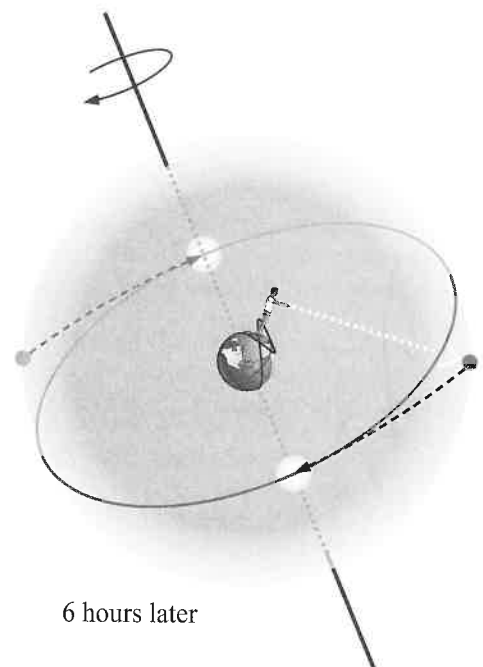
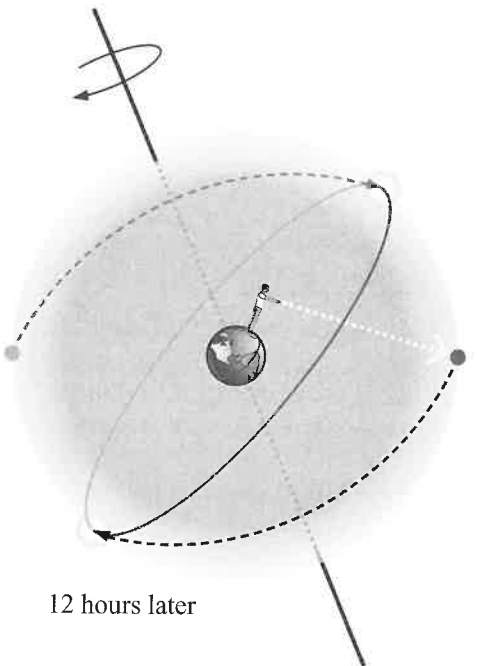


Figure 5a



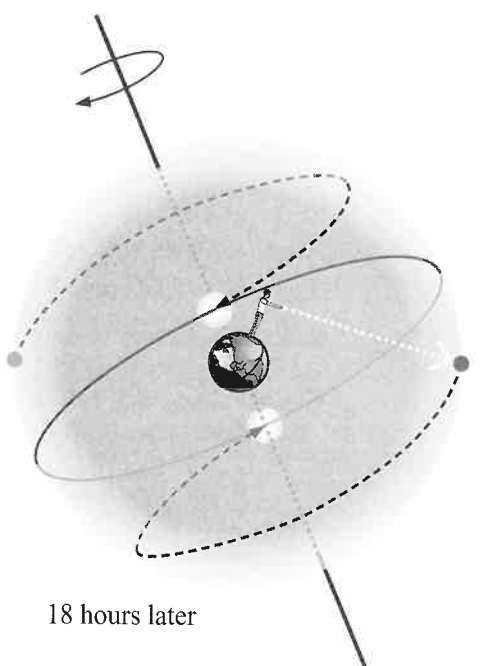
6 hours later

Figure 5b



12 hours later

Figure 5c



18 hours later

Figure 5d